

# Band-Owned Phase Formation and Source-Valid Organization in TGI

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## Abstract

The TGI lattice substrate addresses raw NMU occurrences but does not decide which organizations may become durable knowledge. This paper owns that next step. Eight fast C banks receive adjacency witnesses; eight slow L banks retain temporal coverage through binary carry. Neither path merges, drops, or renames character identities. Source validity is checked from the raw frame, its band ownership, and phase receipt. At the occurrence level, the frozen foundation instance uses  $\Omega_i = (c_{i+1} - c_i)/2$ , threshold one, and a separate lock  $x_i$  conditioned on completeness, address ownership, and uniqueness. At the organization level,  $\Omega_H$  counts distinct unambiguous realizations of raw interval structure and must not be confused with  $\Omega_i$ . Counterevidence can revoke a crystal while retaining the original experience. The paper reports the mechanical evidence and the failed-receipt controls separately from functional interpretation. It derives TGI operators from its own goal of source-valid structural memory using the NMU, MULTIPITA, and Emylton principles as constraints, not as VEYRA NYRA implementation templates.

## 1. Problem and contribution

The identity and lattice paper provides ordered atoms and addresses. Without a validity rule, any adjacency could be read as established knowledge simply because it was stored. TGI instead needs an evidence-dependent phase transition and a way to withdraw the reading right after contradiction. The NINMENI MULTIPITA paper [2] supplies the principle that bands own work rather than partitioning identities. Emylton [3] supplies the distinction among organization state, scalar projection, and valid observation. Crystal Memory [4] demonstrates why a memory store is not itself a semantic claim. TGI derives a different, discrete source-evidence mechanism from those principles.

Four contributions are made: (1) event-preserving C/L ownership; (2) an occurrence validity law separate from an organization support law; (3) a replayable phase certificate that keeps negative evidence; and (4) an explicit decision boundary between a structural crystal and a justified output.

## 2. Band ownership without identity pooling

For a frame  $f = (c_0, \dots, c_{T-1})$ , fast bank  $j \bmod 8$  owns the adjacency witness between positions  $j$  and  $j+1$ . Ownership answers *where the check is maintained*, not which characters exist. Every character still has its own static NMU ID and ordered event. The eight slow banks receive temporal coverage through binary carry. A completed block can be incorporated into a larger block while the raw occurrence order and source remain recoverable. An empty bank is not filled with inferred support.

The invariant is stronger than equal output after processing. Given a stored receipt, an independent checker must recompute the expected owner of each adjacency and temporal block from the original frame length and event order. A receipt that attributes a valid-looking value to the wrong bank is invalid. This protects against the common failure in which a plausible scalar is accepted while the mechanism purported to produce it never ran. The C/L organization here is a TGI structural implementation; it is not the trained channel-partitioned MP2 cell of [2].

### 3. Two levels of density and the lock

The foundation occurrence law uses  $c_i, l_i$  in  $\{0, 1\}$  as validity witnesses for the local and temporal paths:

$\Omega_i = (c_i + l_i)/2$ , with  $\Omega_{crit} = 1$ .

An occurrence becomes structurally eligible only if both checks pass. The versioned  $\xi_i$  lock additionally requires a complete frame, an owned coordinate, and a unique occurrence. This is a certificate bit or scalar projection of the checked state, not an independent source of truth. In particular, a high value caused by a normalization invariant is not a health metric and cannot license a world claim.

An organization  $H$  is inferred from multiple raw episodes. Stable intervals remain literal NMU occurrences. Intervals that vary together may become binding axes, but the axis is an organizational relation over occurrences, not a new token identity. An episode supports  $H$  only when its complete raw sequence has exactly one valid interpretation under  $H$ . Let  $D_k(H)$  be the set of distinct realizations of axis  $k$  among such episodes. The frozen candidate measure is  $\Omega_H = \min_k |D_k(H)|$  with threshold three.  $\Omega_H$  and  $\Omega_i$  have different domains and should never be substituted for one another in a derivation or in a plot.

The phase decision is therefore source-bound. A candidate with enough realizations cannot be promoted if a contributing source has a malformed frame, false C/L ownership, invalid receipt, or counterexample. The checker stores the original evidence scope used for the decision, making a later revocation and re-evaluation possible. No negative raw event is deleted to make a crystal appear stable.

### 4. Falsification protocol

The tests challenge four separable properties:

| Challenge                             | Required outcome                          | Why it matters                             |
|---------------------------------------|-------------------------------------------|--------------------------------------------|
| C/L owner altered, output unchanged   | reject receipt                            | output parity alone misses wrong ownership |
| incomplete or fabricated bank support | no crystallization                        | prevents support from an empty path        |
| counterexample after crystal          | revoke affected reading right             | tests knowledge validity through time      |
| old certificate after new evidence    | reject or revalidate against full history | prevents stale confidence                  |

The raw-organization and phase acceptance reports [6,7] carry the first versioned evidence. The source distribution [8] rebuilds and evaluates the phase from extracted archive bytes, rather than relying on a workspace whose history can change. Historical negative audits are retained in the long-form TGI evidence ledger [9]. The experiment classification follows [2]: a structural check, a regression pass, a systems measurement, and a functional capability claim are distinct kinds of evidence.

### 5. Results, interpretation, and successor interfaces

The accepted raw-organization report [6] records nine gate decisions; the phase report [7] records source-valid phase, counterexample, and polarity checks. The source archive [8] passes its material/hash gates and runs both phase and distribution verifiers from extraction. These results establish a reproducible mechanism for source-valid structural promotion and withdrawal on the tested inputs. The acceptance chain [10] also tests the exact route consequence when the source and frame are valid.

The original TGI manifesto [5] writes a general resonance accumulation for  $\Omega$  and an integral expression for  $\xi$ . The equations above are operational laws for this implementation. Matching names or dimensions would not prove mathematical equivalence to the exploratory manifesto expressions. The paper therefore states the implemented operator and its falsifier directly. This choice makes the chain testable: the next manuscript may form relations only from events whose source and phase are certified. It may not relabel the phase scalar as a semantic predicate.

## 6. Reproducibility and artifacts

The source entry points are *tgi/resonance.py*, *tgi/organization.py*, *tgi/phase\_evidence.py*, and their checkers and tests in the source archive. The frozen contracts [11,12] specify the frame and receipt rules. Raw JSON reports [6–8] and the V71R manifest bind exact bytes and preserve failure cases. No model optimizer, continuous embedding, softmax attention, or subword tokenizer is part of this mechanism. The four NINMENI PDFs [1–4] serve as a derivation reference and remain separate from TGI source code.

## 7. Phase verification algorithm and failure analysis

The independent phase check can be described without relying on a particular Python data structure. It first decodes the declared raw frame and rejects invalid scalar identities. It then reconstructs every fast-bank owner from its adjacency index and every slow-bank block from chronological binary carry. It compares these reconstructed structures with the stored receipt. Only after this equality holds does it compute occurrence validity and the candidate lock. At the organization level it enumerates source episodes, requires complete unambiguous interpretation, derives the set of distinct axis realizations, and checks the current source-validity state. A later counterexample forces a new evaluation over the preserved event history.

This order matters causally. If the checker read an asserted *xi* first and used it to excuse a malformed C/L record, it would make the claimed mechanism self-certifying. If it counted every episode that resembles an axis but has two valid interpretations, *Omega\_H* would be inflated by an undecided source. If it removed an opposing source, the subsequent crystal would appear stronger only because evidence was destroyed. The tested implementation rejects those routes and exposes why promotion is withheld.

| Evidence object     | Unit of observation                    | What it may justify              | What it cannot justify alone     |
|---------------------|----------------------------------------|----------------------------------|----------------------------------|
| <i>c_i,l_i</i>      | one occurrence check                   | local and temporal ownership     | factual truth of text            |
| <i>Omega_i,xi_i</i> | source/frame phase receipt             | structurally readable occurrence | organization across episodes     |
| <i>Omega_H</i>      | distinct unambiguous axis realizations | candidate organization           | correct world referent           |
| counterexample      | later original source event            | revocation or context refinement | deletion of older source history |

The table also prevents a nomenclature failure. NINMENI's Emylton state uses *Omega* for organization history and *xi* as a projected observation; TGI's occurrence and organization receipts use similar letters but own different objects. The shared principle is that organization and projection must be separated. Equality of symbols is not a derivation from Emylton.

## 8. What the results test

The organization report's nine gates and the phase report's six checks are meaningful only with their input scope, exact source bytes, and negative controls. The source-distribution verifier adds a different guarantee: it checks that the same files appear inside the archive with the expected hashes, then executes phase and distribution checks from the extracted tree. It cannot by itself enlarge the functional input class beyond the declared contracts. The functional consequence relevant to this paper is narrower and constructive: a route can be read only through source-valid crystals; when a counterexample revokes that source or organization, the old certificate cannot continue to authorize resolution. The actual query-level effect is checked by the separate resolution manuscript.

## Code and artifact availability

The public TGI runtime and selected regression tests are available at <https://github.com/rfls-emyton/TGI/tree/de95c4c> and as the *tgi-foundation* version *0.2.0.dev0* distribution at <https://pypi.org/project/tgi-foundation/0.2.0.dev0/>. The GitHub commit fixes the code revision used for this publication series. Paths under *evidence/*, the historical *KONSEP.txt* and *HIPOTESIS* archive, and other unpublished workspace reports identify research provenance retained by the author; they are not files in the public GitHub repository or PyPI distribution. Results requiring those receipts are therefore identified by their archived source paths rather than presented as independently downloadable from the code release.

## References

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